



Locating roots by a change of sign

- 1 Show that $f(x) = x^3 - 4x - 2$ has a root between $x = 2$ and $x = 3$.
 - 2 Show that $f(x) = x^3 - x - 1$ has a root between $x = 1$ and $x = 2$.
 - 3 Show that $f(x) = \ln x - 2 + x$ has a root between $x = 1$ and $x = 2$.
 - 4 Show that $f(x) = x^3 - 2x - 5$ has a root between $x = 2$ and $x = 3$.
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Interval bisection

- 5 Using $f(x) = x^3 - x - 1$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval containing the root.
 - 6 Continuing from the previous problem, evaluate $f(1.25)$ and give a narrower interval for the root.
 - 7 Using $f(x) = \ln x - 2 + x$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval for the root.
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Fixed-point iteration

- 8 The equation $x^3 - 4x - 2 = 0$ rearranges to $x = \sqrt[3]{4x + 2}$. Using $x_0 = 2$, find x_1 , x_2 , and x_3 , to four decimal places.
 - 9 The equation $x^2 - 5x + 2 = 0$ rearranges to $x = 5 - \frac{2}{x}$. Using $x_0 = 4$, find x_1 and x_2 , to four decimal places.
 - 10 The equation $x^3 - 2x - 5 = 0$ rearranges to $x = \sqrt[3]{2x + 5}$. Using $x_0 = 2$, find x_1 , to four decimal places.
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Convergence and divergence of iterations

- 11 Explain why the iteration $x_{n+1} = x_n^3 - 3$ (i.e. $g(x) = x^3 - 3$), starting from $x_0 = 1.5$, would fail to converge to a root of $x^3 - x - 3 = 0$ near $x = 1.5$.
- 12 Using $g(x) = 5 - \frac{2}{x}$ from the earlier iteration, find $g'(x)$ and evaluate $|g'(x)|$ at $x = 4.56$ (near the root), to explain why the iteration converges there.

Rearranging equations for iteration

- 13 Rearrange $x^2 - 3x - 1 = 0$ into the form $x = g(x)$ by making the linear term the subject.
- 14 Rearrange $x^3 + x - 1 = 0$ into the form $x = \frac{1}{x^2 + 1}$, and verify algebraically that this is equivalent to the original equation.

The Newton-Raphson method

- 15 Use one iteration of the Newton-Raphson method, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 4x - 2$, to four decimal places.
- 16 Continuing from $x_1 = 2.25$ in the previous problem, find x_2 , to four decimal places.
- 17 Use the Newton-Raphson method with $x_0 = 2.5$ to find x_1 for $f(x) = x^2 - 7$ (approximating $\sqrt{7}$), to four decimal places.
- 18 Use the Newton-Raphson method with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 2x - 5$, to four decimal places.

Further Newton-Raphson: rapid convergence and failure cases

- 19 Using $x_1 = 2.65$ from the previous problem for $f(x) = x^2 - 7$, perform a second iteration to find x_2 , to four decimal places, and compare with $\sqrt{7} \approx 2.6458$.
- 20 Explain why $x_0 = 1$ is an unsuitable starting point for the Newton-Raphson method applied to $f(x) = x^3 - 3x + 1$.
- 21 Continuing from $x_1 = 2.1$ for $f(x) = x^3 - 2x - 5$, perform a second Newton-Raphson iteration to find x_2 , to four decimal places.

Error bounds from bisection

- 22 A root of $f(x) = x^3 - 4x - 2$ lies in $(2.21, 2.22)$, since $f(2.21) \approx -0.046$ and $f(2.22) \approx 0.061$. State the midpoint estimate and the maximum possible error.
- 23 Given that the root of $f(x) = x^3 - 4x - 2$ is trapped in the interval $(2.2141, 2.2144)$ after further bisection, state the root correct to three decimal places, justifying your answer.

- 24 A root of $f(x) = x^3 - 2x - 5$ lies in $(2.09, 2.10)$, since $f(2.09) \approx -0.051$ and $f(2.10) \approx 0.061$. State the midpoint estimate and the maximum possible error.
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A well-designed word problem: designing a support strut

- 25 This question has four parts. An engineer models the cross-sectional profile of a support strut by $f(x) = x^3 - 6x^2 + 9x - 2$ (metres) and needs to find where it crosses the x -axis. (a) Show by a change of sign that a root lies between $x = 0$ and $x = 1$. (b) Use the Newton-Raphson method with $x_0 = 0.5$ to find x_1 , to four decimal places. (c) Perform a second iteration to find x_2 , to four decimal places. (d) Perform a third iteration to find x_3 , and state the root to three decimal places.